

**Orders management system (OMS)
SRS document**

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1. Introduction

1.1 Purpose

The GIU Food Ordering System is a web-based application designed to streamline food ordering on the German International University (GIU) campus.

The system allows GIU community members to reserve menu items from approved on-campus food vendors, so their meals are prepared and ready between classes, reducing waiting times during peak periods.

Vendors on campus use the system to manage incoming orders, update food preparation status.

The application gives food truck customers flexible control of their pickup times which eliminates the problem of long waiting lines while providing vendors with efficient order management

1.2 Intended Audience

- A. Students, Staff, Faculty members, visitors : can browse the menus, schedule their orders pick-up time, keep track of their orders
- B. Vendors (food trucks): Receive orders, manage menus, confirm pickups

1.3 Scope

A. The product provides:

a. Student features:

Sign up as a customer, Login using user name and password, schedule order, menus browsing, Browsing food trucks traffic and availability, menu ordering with estimated pickup times, order tracking, immediate pick-up for the cancelled orders from other customers, ability to delay or cancel order if the order is not being prepared.

b. Vendor features:

Sign up as food truck service provider, log in with user name and password, menu management with prep times, ability to update order status(Pending, In Preparation, Ready), ability to update the food truck status (very busy, neutral, not busy),ability to enter the next time available time slot.

B. The product excludes:

a. Online payments (only cash on pickup).

b. Third-party delivery or off-campus vendor access.

2. Product Vision and Scope

A. Vision :

“To make food ordering at GIU as seamless as attending a lecture—students order during class, vendors prepare on time, pickups are instant and no waiting lines.”

B. The product will:

- a. Enable class-schedule-driven ordering with estimated pickup flexibility.
- b. Provide cash-on-pickup transactions only.
- c. Consolidate all GIU food vendors in a unified platform.
- d. Show which vendors are open/available at any given time.
- e. Show the next available time slot for ordering from busy trucks.
- f. Provide a menu for immediate ready to pick up order.
- g. Provide the option for browsing menus and making new order .

C. The product will not:

- a. Process online payments.
- b. Offer external delivery or off-campus access.

3. High-Level Architecture

3.1 Conceptual Design

- A. Frontend: Built with Modern HTML, CSS, JavaScript, and JQuery.
Or JavaScript Frame-work for responsive web and mobile experience.
- B. Backend: RESTful API using [Node.js](#)
- C. Database: PostgreSQL managed with Prisma
- D. Order Management: Orders stored and tracked in PostgreSQL;
real-time updates delivered via backend polling or WebSockets
- E. Vendor Interface: Role-based vendor portal inside platform for
menu editing, managing orders,

3.2 Technology Stack

- A. Frontend: Modern HTML, CSS, JavaScript, and JQuery. Or JavaScript Frame-work
- B. Backend: RESTful API using [Node.js](#)
- C. Database: Relational (PostgreSQL)
- D. Authentication: currently, none
- E. Payment Gateway: currently, none
- F. Hosting: currently, none

4. Functional Requirements

4.1 Student Features (User Stories)

As a student, I can sign up as a customer for the first time and then log in to my account using my username and password. I can browse all food vendors with menus and availability status in real time and schedule my order. If time is tight, I can immediately pick up a ready order cancelled by someone else. I can order an item in advance and get an estimated pickup time window. I can track my order status (Pending, In Preparation, Ready, no longer available). I can delay or cancel my order if it is not in preparation. I can pick-up my order and pay cash on site. If I didn't show up within 15 minutes my order is going to be cancelled and resold on the immediate pick-up menu.

4.2 Vendor Features (User Stories)

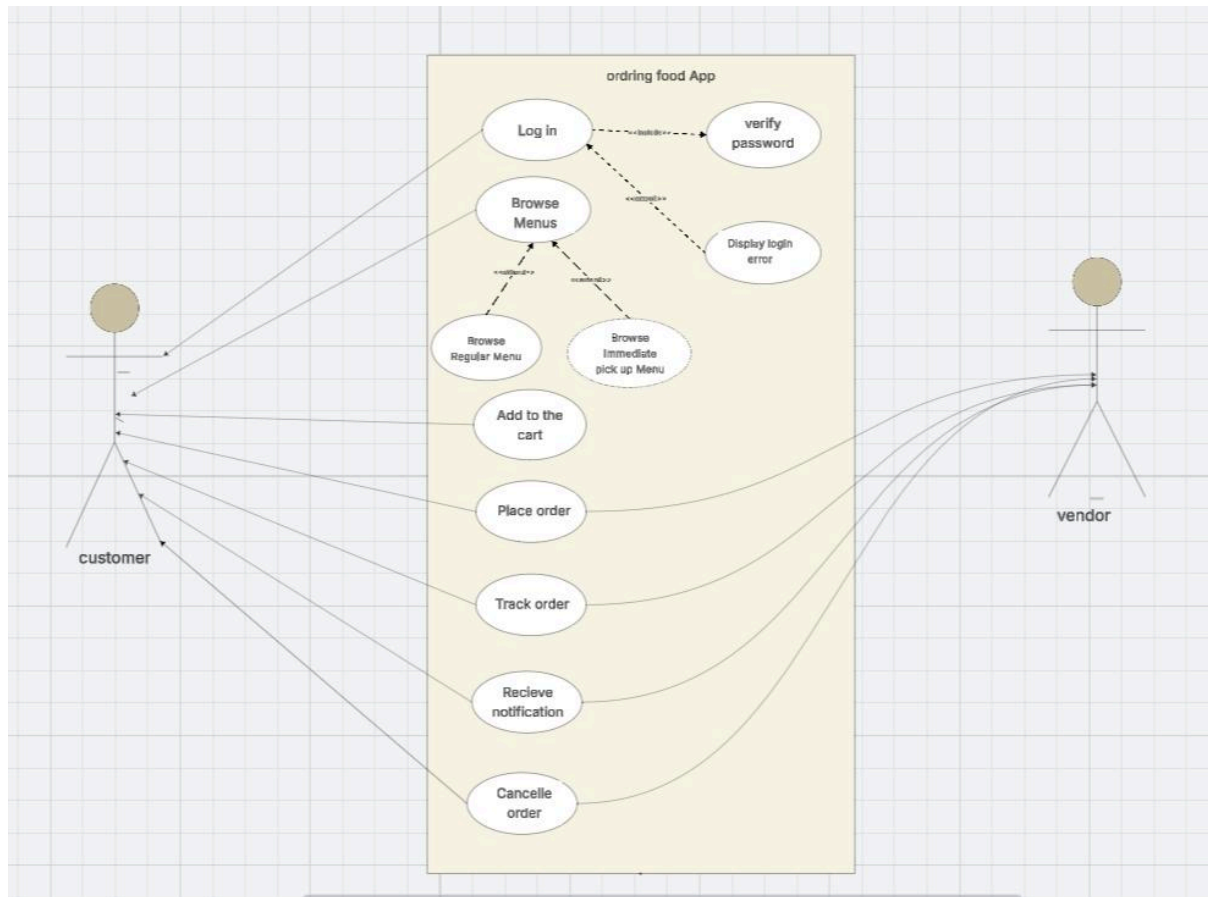
As a vendor, I can sign up for the first time as a service provider and log in with my username and password. I can set the food truck traffic status (very busy, neutral, not busy) and the next time slot available for ordering. I can manage my menu (items, prices, estimated prep times). I can track incoming orders and update the orders status (Pending, In Preparation, Ready) in real-time. I can push the ready-orders that the customers cancelled or didn't show up (after 15 minutes from the pick-up time had passed) on the immediate pick-up menu.

4.4 Use Case Diagram

A. Actors: Student, Vendor.

B. Use Cases:

- a. Students: Login, Set Schedule, Browse Vendors, Place Order, Track Order, see cancelled order, delay order, cancelled order.
- b. Vendors: Signup, Manage Menu, Manage Hours, Update Order Status.



5. Non-Functional Requirements

- A. Performance: Handle up to 500 concurrent users.
- B. Response Time: 2–3 seconds per request.
- C. Security: rate limiting and input validation
- D. Reliability: $\geq 99\%$ availability during 9 AM–5 PM for community members and from 8:30 till 5:30 for vendors.
- E. Scalability: Must support all GIU vendors and community members .
- F. Usability: Mobile-first design, intuitive schedule setup (table toggles).
- G. Backup/Recovery: Daily automatic backups of databases.

6. Interface Requirements

At this stage:

- A. Web app (GIU community members, vendors,) only
- B. Vendor system integrated into the same platform via role differentiation.

7. Constraints

- A. Operating Hours: 9:00 AM to 5:00 PM daily.
- B. Vendors set their own availability windows, shown as “online/offline.
- C. Delivery Limitation: Pickup-only, no delivery.
- D. Payments: Cash-only transaction.
- E. Time Constraint: Must be developed by the end of semester.

8.Future Scope

- A. Show a welcome message when the user opens the app.
- B. Allow the customer to rate the restaurant after receiving the order (stars + optional comment).
- C. Allow the restaurant to rate customers and decline orders from customers that have a history of not showing up
- D. Let the vendor send a notification to the customer when the order is ready.
- E. User verification as GIU community member
- F. Credit card authentication service
- G. Authentication of food truck service providers
- H. Vendors can update their items stock status (available, out of stock)